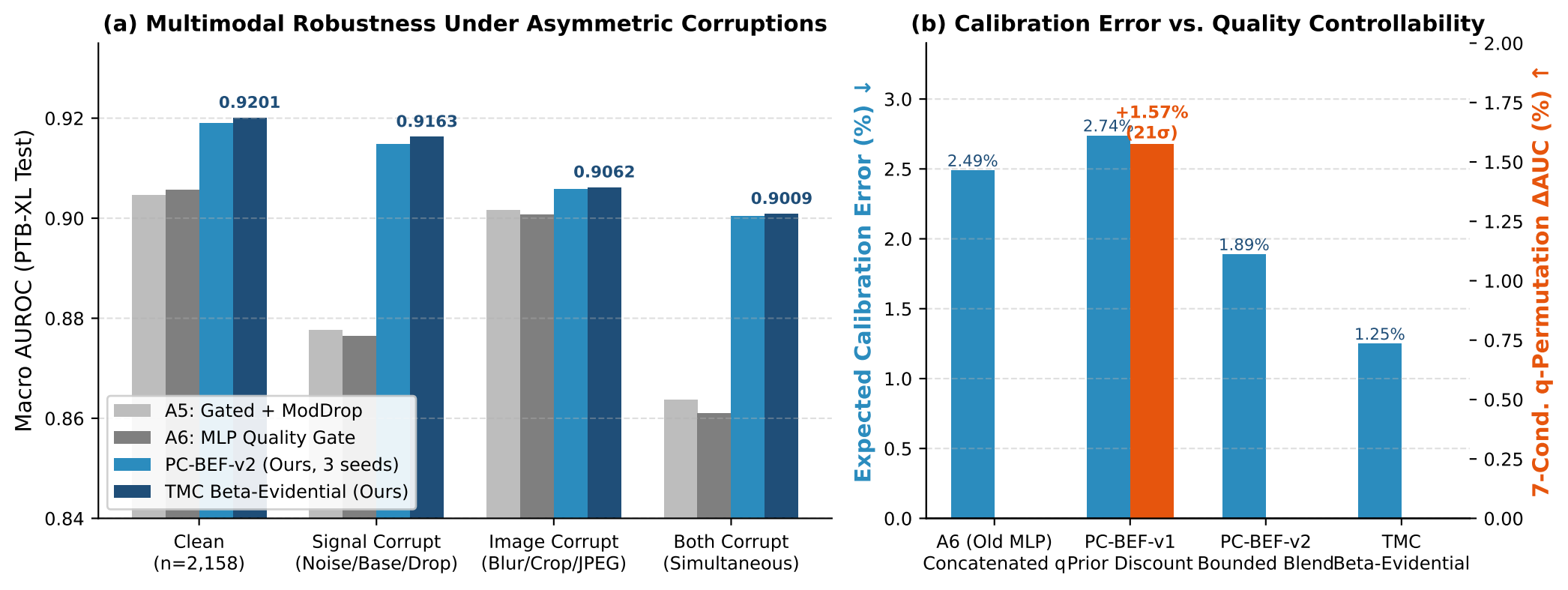
(a) Multimodal Robustness Under Asymmetric Corruptions
Macro AUROC (PTB-XL Test)
0.92
0.90
0.88
0.86
0.84
0.9201
0.9163
0.9062
0.9009
A5: Gated + ModDrop
A6: MLP Quality Gate
PC-BEF-v2 (Ours, 3 seeds)
TMC Beta-Evidential (Ours)
Clean
(n=2,158)
Signal Corrupt
(Noise/Base/Drop)
Image Corrupt
(Blur/Crop/JPEG)
Both Corrupt
(Simultaneous)
(b) Calibration Error vs. Quality Controllability
Expected Calibration Error (%) ↓
7-Cond. q-Permutation ΔAUC (%) ↑
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
2.49%
2.74%
+1.57%
(21σ)
1.89%
1.25%
A6 (Old MLP)
Concatenated q
PC-BEF-v1
Prior Discount
PC-BEF-v2
Bounded Blend
TMC
Beta-Evidential